Mandatory exercise 3

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Question 1

Consider the basis for P_2 given by $B = \{p, q, r\}$, where

$$p = 3x^2 - 2 \qquad q = 2x^2 + x \qquad r = x - 1$$

Compute the coordinates of the polynomial $s = 10x^2 + x - 10$ relative to B.

Computing coordinates relative to basis

By definition:

$$[s]_B = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

Therefore:

$$c_1(3x^2 - 2) + c_2(2x^2 + x) + c_3(x - 1) = 10x^2 + x - 10$$

Written as a system of linear equations:

$$\begin{cases} 3c_1 + 2c_2 = 10 \\ c_2 + c_3 = 1 \\ -2c_1 - c_3 = -10 \end{cases}$$

Solve as a matrix:

$$\begin{bmatrix} 3 & 2 & 0 \\ 0 & 1 & 1 \\ -2 & 0 & -1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 1 \\ -10 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 3 & 2 & 0 & 10 \\ 0 & 1 & 1 & 1 \\ -2 & 0 & -1 & -10 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1 + R_3} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & 1 & 1 \\ -2 & 0 & -1 & -10 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow R_3 + 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 4 & -3 & -10 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow R_3 - 4R_2} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & -7 & -14 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow -\frac{1}{7}R_3} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$\xrightarrow{R_2 \leftarrow R_2 - R_3} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_2 + R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

Concluding:

$$[s]_B = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}$$

Question 2

The aim of this exercise is to find the linear combination of the functions

$$f(x) = 2x^2 + x \qquad\qquad g(x) = x + 1$$

that best fits the observation points $\{(-1, -2), (0, 2), (1, 5)\}$. You should do this by answering the following questions

(a) Construct a basis for the following subspace of $\mathbb{R}^3$

$$V = \{(h(-1), h(0), h(1)) \mid h(x) = a \cdot f(x) + b \cdot g(x) \text{ for some } a \text{ and } b\}$$

Let $B = \{u, v\}$ be the basis of V, where:

$$u = (f(-1), f(0), f(1)) = (1, 0, 3)$$

$$v = (g(-1), g(0), g(1)) = (0, 1, 2)$$

(b) Construct a projection matrix for V

First construct the matrix, where the columns are the basis vectors of V.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 3 & 2 \end{bmatrix}$$

The projection matrix is calculated as:

$$
\begin{aligned}
P &= A(A^TA)^{-1}A^T \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 3 & 2 \end{bmatrix} \left(\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 3 & 2 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 3 & 2 \end{bmatrix} \left(\begin{bmatrix} 10 & 6 \\ 6 & 5 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix} \\
&= \frac{1}{14} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 5 & -6 \\ -6 & 10 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix} \\
&= \frac{1}{14} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 5 & -6 & 3 \\ -6 & 10 & 2 \end{bmatrix} \\
&= \frac{1}{14} \begin{bmatrix} 5 & -6 & 3 \\ -6 & 10 & 2 \\ 3 & 2 & 13 \end{bmatrix}
\end{aligned}
$$

(c) Use projection to compute the numbers a, b, such that $h(x) = a \cdot f(x) + b \cdot g(x)$ minimises the sum

$$e^2 = (h(-1) + 2)^2 + (h(0) - 2)^2 + (h(1) - 5)^2$$

Let:

$$x = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \qquad\qquad y = \begin{bmatrix} -2 \\ 2 \\ 5 \end{bmatrix}$$

Normally we would calculate a and b directly by multiplying $(A^TA)^{-1}A^Ty$. Since we already have the full projection matrix, we can find the predicted y values that minimise the squared errors and use them to find the function that generates them.

$$h = \begin{bmatrix} h(-1) \\ h(0) \\ h(1) \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 5 & -6 & 3 \\ -6 & 10 & 2 \\ 3 & 2 & 13 \end{bmatrix} \begin{bmatrix} -2 \\ 2 \\ 5 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ 3 \\ \frac{9}{2} \end{bmatrix}$$

We use the first 2 values in x and h vectors to calculate a and b and use the 3rd one to validate the result.

$$
\begin{aligned}
h(x) &= a(2x^2 + x) + b(x + 1) \\
-\frac{1}{2} &= a(2 - 1) + b(-1 + 1) \\
a &= -\frac{1}{2} \\
3 &= a(0) + b(0 + 1) \\
b &= 3
\end{aligned}
$$

Validate 3rd pair:

$$\begin{aligned}\frac{9}{2} &= -\frac{1}{2}(2+1) + 3(1+1)\\ \frac{9}{2} &= -\frac{3}{2} + 6\\ \frac{12}{2} &= 6\\ 0 &= 0\end{aligned}$$

Therefore $a = -\frac{1}{2}$ and $b = 3$ minimise the sum of the squares of the errors.

Explain in your own words which point you have projected to V, what the numbers a and b have to do with this projection, and why the projection solves the problem of minimising the sum e^2 above.

We have projected the point $y = \begin{bmatrix}-2\\2\\5\end{bmatrix}$ to V, where the point closest to y is h. The numbers a and b are the coordinates of point h in the basis B. This minimises e^2 because the distance between points y and h is calculated with the application of the pythagorean theorem in 3D.

$$\begin{aligned}\text{distance} &= \sqrt{(\text{distance in 1st coordinate})^2 + (\text{distance in 2nd coordinate})^2 + (\text{distance in 3rd coordinate})^2}\\ &= \sqrt{(h(-1)+2)^2 + (h(0)-2)^2 + (h(1)-5)^2}\\ \text{distance}^2 &= (h(-1)+2)^2 + (h(0)-2)^2 + (h(1)-5)^2\\ \text{distance}^2 &= e^2\end{aligned}$$